

What is CV?

↳ is a subset of AI where we provide ability to machine to process visual info like image, video etc.

Applications of CV:

① Image classification



② Object Recognition



③ Image segmentation

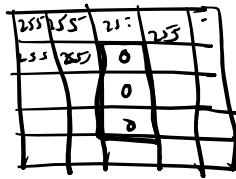
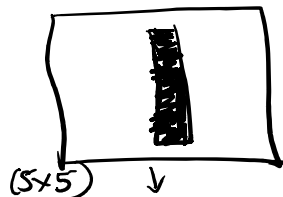


CNN → convolutional Neural Network is an algorithm introduced to specifically work on visual data.

What is digital image? → is a collection of pixels (number which represents colors → 0 to 255) (2D representation)

① Black white image (Grey Scale)

0 → black
255 → white



② RGB scale image

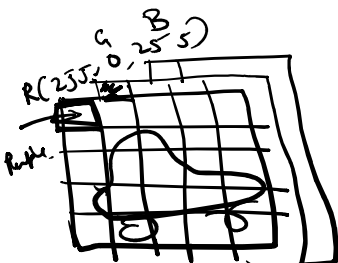
0 to 255 for R, G, B

(0, 0, 0) → Black image

(255, 0, 0) → Red image

(0, 255, 0) → Green

(0, 0, 255) → Blue





2 yr

CNN

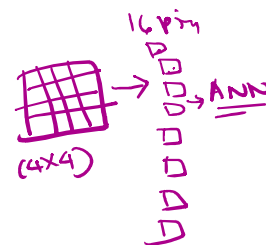
Occipital lobe in brain has different layers. Each layer is responsible for different task.

One layer → identify shape
 another layer → identify edges/corner.
 another layer → identify objects within an image.

Using this working on Occipital lobe, CNN was introduced.



ANN Disadv
visual image process

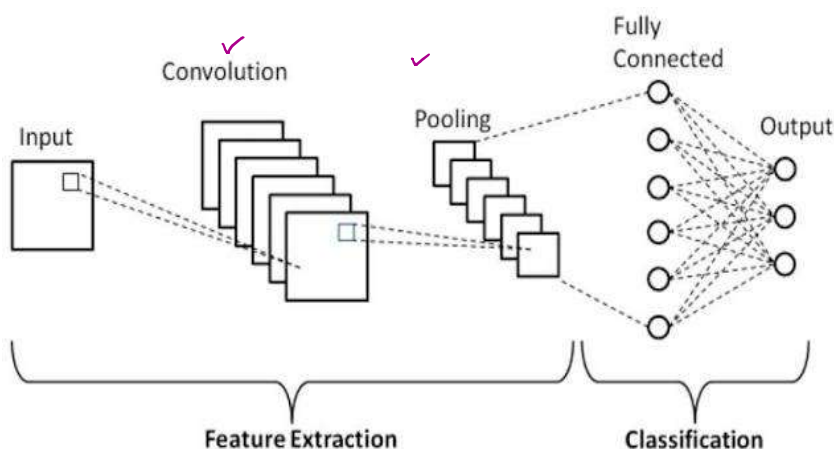


① Lot of training data has to be provided. Fails to identify different orientations things

② ANN requiring 2D images to be flatten to 1D pixel repr. & during training it fails to identify relationships between independent pixel.

③ Computational time & cost is higher with ANN.

CNN architecture

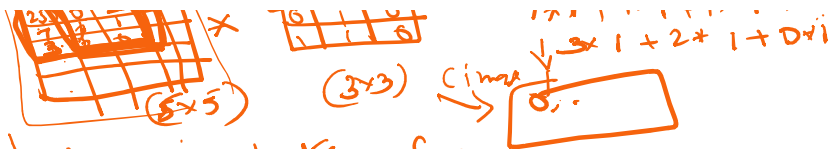


① Convolution layer → where we extract imp feature from an image by performing

convolution operation on filters (kernel).

$$\begin{array}{|c|c|c|c|} \hline 2 & 0 & 2 & 1 \\ \hline 2 & 3 & 1 & 1 \\ \hline 3 & 1 & 1 & 0 \\ \hline \end{array} \times \begin{array}{|c|c|c|} \hline 1 & 0 & 1 \\ \hline 0 & 1 & 0 \\ \hline 1 & 1 & 0 \\ \hline \end{array} = \begin{array}{l} (2 \times 5 \times 1 + 2 \times 3 \times 1 + 2 \times 1) + \\ 7 \times 1 + 1 \times 1 + 1 \times 1 + \\ 3 \times 1 + 2 \times 1 + 0 \times 1 \end{array}$$

(3x3) (image)



Stride → step size taken for matrix multiplication

Padding → adding extra layer to the image to avoid shrinking of input image when its multiplied by filter.



3	5	1
3	6	1
4	7	9

No padding

0	0	0	0	0
0	3	5	1	0
0	3	6	1	0
0	4	7	9	0
0	0	0	0	0

Zero padding

3	3	5	1	1
3	3	5	1	1
3	3	6	1	1
4	4	7	9	9
4	4	7	9	9

Replication padding

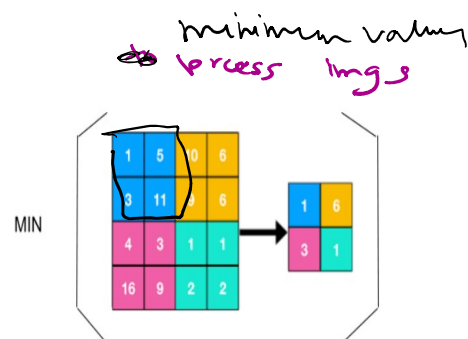
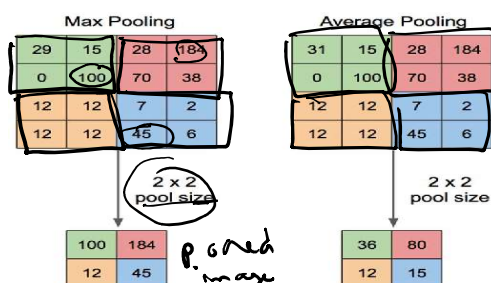
Pooling layer:

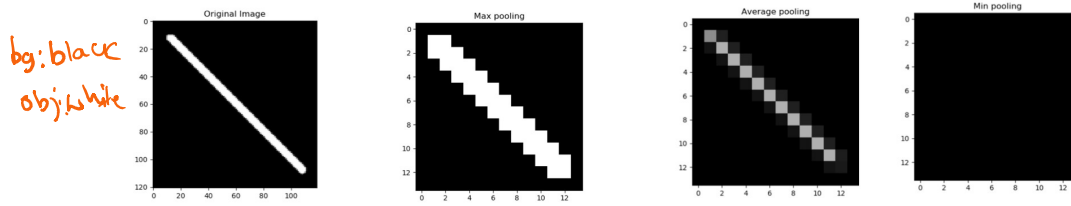
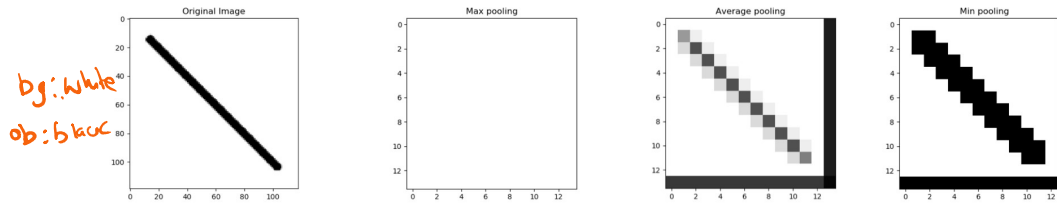
3x3x6

o/p of convolution layer i.e feature map is passed through pooling layer.

Here we down sample the image.

↑
reduce size/dimension





process imgs

- * Min pools works better when object is darker & background is brighter.
- * max pool works better when object is brighter & background is darker.

